

Structural, Electrical, and Magnetic Versatility of the Oxalate-Based [CuFe] Compounds Containing 2,2':6',2''-Terpyridine: Anion-Directed Synthesis

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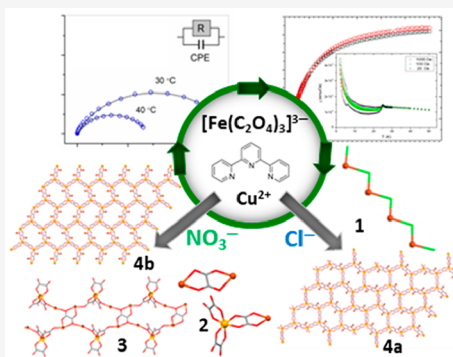
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ABSTRACT: The heterodimetallic [CuFe] compounds $[\text{Cu}^{\text{II}}(\text{terpy})_4\text{Cl}_5][\text{Fe}^{\text{III}}(\text{C}_2\text{O}_4)_3] \cdot 10\text{H}_2\text{O}$ (**1**; terpy = 2,2':6',2''-terpyridine), $[\text{Cu}^{\text{II}}_2(\text{H}_2\text{O})_2(\text{terpy})_2(\text{C}_2\text{O}_4)] [\text{Cu}^{\text{II}}\text{Fe}^{\text{III}}(\text{CH}_3\text{OH})(\text{terpy})(\text{C}_2\text{O}_4)_3]_2$ (**2**), and $\{[\text{Cu}_2^{\text{II}}\text{Fe}^{\text{III}}(\text{H}_2\text{O})(\text{terpy})_2(\text{C}_2\text{O}_4)_{7/2}] \cdot 6\text{H}_2\text{O}\}_n$ (**3**) were obtained using building block approach, from reaction of aqueous solution of $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and a methanol solution containing Cu^{2+} ions and terpy by the layering technique. Interestingly, by changing only the anion of the starting salt of copper(II), $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ instead of $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$, an unexpected change in the type of bridge, oxalate (**2** and **3**) versus chloride (**1**), was achieved, thus affecting the overall structural architecture. Two polymorphs of 3D coordination polymer $[\text{Cu}^{\text{II}}\text{Fe}^{\text{II}}_2(\text{H}_2\text{O})(\text{terpy})(\text{C}_2\text{O}_4)_3]_n$ (**4**), crystallizing in the triclinic (**a**) and monoclinic (**b**) space groups, were formed hydrothermally, depending on whether $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ or $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ was added to the water, besides $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ and terpy, respectively. Under hydrothermal conditions iron(III) from initial building block is reduced to the divalent state, creating 2D honeycomb $[\text{Fe}^{\text{II}}_2(\text{C}_2\text{O}_4)_3]^{2n-}$ layers, which are bridged by $[\text{Cu}(\text{H}_2\text{O})(\text{terpy})]^{2+}$ cations. Compounds were investigated by single-crystal X-ray diffraction, IR, and impedance spectroscopies, magnetization measurements, and density functional theory (DFT) calculations. In compounds **1** and **2**, 0D magnetism is observed, with **1** having a ground-state spin of 1 due to different interactions through chloride bridges of Cu^{2+} ions in tetramer $[\text{Cu}_4^{\text{II}}(\text{terpy})_4\text{Cl}_5]^{3+}$ and **2** showing strong antiferromagnetic coupling of Cu^{2+} ions mediated by oxalate ligand in $[\text{Cu}_2^{\text{II}}(\text{H}_2\text{O})_2(\text{terpy})_2(\text{C}_2\text{O}_4)]^{2+}$ and weak ones between Cu^{2+} and Fe^{3+} ions through oxalate bridge in $[\text{Cu}^{\text{II}}\text{Fe}^{\text{III}}(\text{CH}_3\text{OH})(\text{terpy})(\text{C}_2\text{O}_4)_3]^-$. Polymer **4** exhibits antiferromagnetic phase transition at 25 K: The $[\text{Fe}^{\text{II}}_2(\text{C}_2\text{O}_4)_3]^{2n-}$ layers are antiferromagnetically ordered, and a small amount of interlayer interaction is transferred through $[\text{Cu}(\text{H}_2\text{O})(\text{terpy})]^{2+}$ cations via $\text{O}_{\text{ox}}-\text{Cu}-\text{O}_{\text{ox}}$ bridges. Additionally, compounds **1** and **2** are electrical insulators, while **4a** and **4b** show proton conductivity.



INTRODUCTION

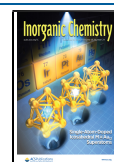
Due to the rich variety of metal species and ligands, coordination geometry, guests inside the pores, and supra-molecular structures, an enormous number of coordination polymers with various structures, dimensionality, and nuclearity have been synthesized and reported.¹ This area of molecular chemistry is the basis for design of the molecule-based materials that combine two (or more) physical properties of interest, especially upgrading and expanding the molecular magnetism toward multifunctional systems.

In this context, the oxalate-based compounds of various transition metals have been the subject of numerous studies; the oxalate moiety, $\text{C}_2\text{O}_4^{2-}$ (ox), acts as a linker between metal centers, with various possibilities of coordination to metal centers yielding open structures with dimensionalities ranging from 0 to 3. A property of oxalate ions that has been of specific interest is their ability to mediate electronic effects between paramagnetic metal ions.

The combination of magnetic oxalate-based coordination polymers with organic/inorganic functional cations provides a suitable approach for designing hybrid magnetic materials exhibiting multifunctional properties.^{2–4} Most of the oxalate-based molecular magnets described to date have been obtained by the “complex-as-ligand approach”. In this synthetic strategy, a molecular building block, the tris(oxalato)metalate $[\text{M}^{\text{III}}(\text{C}_2\text{O}_4)_3]^{3-}$ anion ($\text{M}^{\text{III}} = \text{Cr}, \text{Fe}, \text{Ru}, \text{Rh}, \text{Mn}, \text{or V}$), is used as a ligand toward other metal cations.¹ The topology of these oxalate-bridged compounds is controlled by a templating counterion. Therefore, layered 2D honeycomb structures of

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the formulas $[M^{II}_2(C_2O_4)_3]_n^{2n-}$ and $[M^{II}M^{III}(C_2O_4)_3]_n^{n-}$, showing ferro-, ferri-, or canted antiferromagnetic ordering have been obtained using a bulky charge-compensating molecular cation. Magnetic multilayers, photochromic magnets, ferromagnetic molecular metals, or chiral magnets could be formed by combining the cooperative magnetism of the oxalate systems and the electronic properties supplied by the cationic molecular lattice, as using paramagnetic decamethylferrocenium or organic radical cations, photochromic molecules, nonlinear optical (NLO)-active molecules, organic *p*-electron donors, and chiral cations. Other magnetic networks, different from the 2D honeycomb-like network, can also be obtained depending on the nature of the templating cation (size, shape, and charge). The use of the tris-chelated $[M(bpy)_3]^{2+}$ or $[M(bpy)_3]^{3+}$ (*bpy* = 2,2'-bipyridine) entities provides the family of 3D networks of the formulas $[M^{II}_2(C_2O_4)_3]_n^{2n-}$, $[M^{II}M^{III}(C_2O_4)_3]_n^{n-}$, and $[M^{II}M^{III}(C_2O_4)_3]_n^{n-}$. The homochiral 3D anionic networks have lower critical temperatures than their heterochiral 2D analogues, although the pathway for magnetic interaction is mainly the same, due to different relative orientation of the magnetic orbitals and to longer metal-to-metal distances.^{1,2} Several compounds were reported which were formed by inserting the complexes of Fe^{III} , as $[Fe(sal_2-trien)]^+$ or $[Fe(bpp)_2]^{2+}$ ($H_2sal_2-trien$ = *N,N'*-disalicylidene-*tri*ethylene-tetramine; *bpp* = 2,6(bispyrazol-3-yl)pyridine), between 2D $[Mn_2^{II}(C_2O_4)_3]_n^{2n-}$ and $[Mn^{II}Cr^{III}(C_2O_4)_3]_n^{n-}$ layers, or by encapsulating in a 3D $[Mn^{II}Cr^{III}(C_2O_4)_3]_n^{n-}$ network, in order to afford materials with coexistence of magnetic ordering and spin-crossover, in which the critical temperature of the magnet can be tuned by applying an external stimulus (light and pressure) acting on the spin-crossover component.^{5–10}

Hydrothermal reactions have been extensively applied to the synthesis of inorganic–organic hybrid materials giving structural novelty and unique properties. It has been shown that hydrothermal conditions often produce (meta)stable compounds that may otherwise not be accessible by conventional methods. Coordination polymers are synthesized by combining organic ligands and metal salts in solvothermal reactions at relatively low temperatures (below 300 °C). The most important parameters of this synthesis which affect the topology of the networks are temperature, concentrations of metal salt and ligand (which can be varied across a large range), the extent of solubility of the reactants in the solvent, the rate of crystallization, and the pH of the solution. This method also gives precise control over the size, shape, distribution, and crystallinity of metal–organic materials.^{11–13}

In our previous works we reported structural and magnetic properties of the heterometallic oxalate-bridged coordination polymers of different dimensionality obtained using $[M^{III}(C_2O_4)_3]^{3-}$ (M^{III} = Cr, Mn, and Fe) as building blocks.^{14–17} Following these previous investigations and obtained cognition, we have now focused on the preparation of novel [CuFe] compounds by the layering technique and under the hydrothermal conditions using $[Fe^{III}(C_2O_4)_3]^{3-}$ as precursor and tridentate 2,2':6',2''-terpyridine (*terpy*) ligand, unlike our previous studies in which we exploited bidentate *N*-donor ligands, mostly 2,2'-bipyridine and 1,10-phenanthroline. It is known that the use of the capping ligands, in addition to the stabilization of the solid-state structures, can control and influence the dimensionality of coordination systems.²

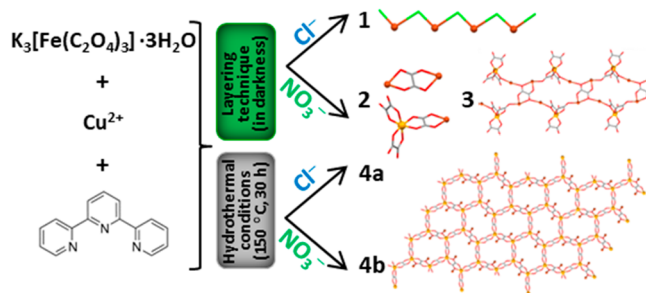
In this publication, we report structural diversity reflected on magnetic and electrical properties of novel oxalate-based

[CuFe] compounds containing *terpy* as *N*-donor ligand: monochloride- and oxalate-bridged complex salts $[Cu^{II}_4(terpy)_4Cl_5][Fe^{III}(C_2O_4)_3] \cdot 10H_2O$ (1) and $[Cu^{II}_2(H_2O)_2(terpy)_2(C_2O_4)][Cu^{II}Fe^{III}(CH_3OH)(terpy)(C_2O_4)_3]_2$ (2), respectively, and two polymorphs of compound $[Cu^{II}Fe^{II}_2(H_2O)(terpy)(C_2O_4)_3]_n$ (4), containing well-known 2D honeycomb layers $[Fe^{II}_2(C_2O_4)_3]_n^{2n-}$, which are bridged by paramagnetic $[Cu(H_2O)(terpy)]^{2+}$ cations, ultimately giving 3D network showing long-range magnetic order. Additionally, the resulting 1D oxalate-bridged ladderlike coordination polymer $\{[Cu_2^{II}Fe^{III}(H_2O)(terpy)_2(C_2O_4)_{7/2}] \cdot 6H_2O\}_n$ (3) was also structurally and spectroscopically characterized.

RESULTS AND DISCUSSION

Synthesis. The titled [CuFe] compounds $[Cu^{II}_4(terpy)_4Cl_5][Fe^{III}(C_2O_4)_3] \cdot 10H_2O$ (1), $[Cu^{II}_2(H_2O)_2(terpy)_2(C_2O_4)][Cu^{II}Fe^{III}(CH_3OH)(terpy)(C_2O_4)_3]_2$ (2), and $\{[Cu_2^{II}Fe^{III}(H_2O)(terpy)_2(C_2O_4)_{7/2}] \cdot 6H_2O\}_n$ (3) were obtained using building block approach, from the reaction of aqueous solution of $[Fe(C_2O_4)_3]^{3-}$ and methanol solution containing Cu^{2+} ions and 2,2':6',2''-terpyridine in the molar ratio of 1:1:1, by the layering technique,¹⁸ without presence of the daylight. Fascinatingly, by altering only the anion of the starting salt of copper(II) in the methanol layer, $Cu(NO_3)_2 \cdot 3H_2O$ instead $CuCl_2 \cdot 2H_2O$, an unexpected change in type of the bridge, oxalate versus chloride, was achieved, thus affecting the overall structural architecture (Scheme 1). It appears that due to steric effect

Scheme 1. Schematic Representation of the Synthesis of Compounds 1–4



related to terpyridine molecule, i.e., due to the use of the more bulky tridentate *N*-donor ligand, smaller chloride ions present in the reaction solution kinetically favor formation of the bridge in regard to the oxalate ligand. This is substantiated by the fact that the bridging by chloride did not occur when smaller bidentate 2,2'-bipyridine ligand was used instead of *terpy* species by utilization of the slow liquid diffusion.¹⁷ In a test tube containing nitrate anions, partial decomposition of the tris(oxalato)ferrate(III) anion leads to the release of oxalate group from the coordination sphere of iron(III). During the crystallization process, this oxalate ligand is consequently coordinated to copper(II) ions in reaction mixture, yielding an oxalate-bridged dinuclear $\{[Cu^{II}(H_2O)(terpy)]_2(\mu-C_2O_4)\}^{2+}$ cationic unit of compound 2 or coordination polymer 3 containing oxalate-bridged dinuclear entity of copper(II) $\{[Cu^{II}(terpy)]_2(\mu-C_2O_4)\}^{2+}$ extra linked.^{16,17}

Two polymorphs of 3D coordination polymer $[Cu^{II}Fe^{II}_2(H_2O)(terpy)(C_2O_4)_3]_n$ (4), crystallizing in the triclinic *P* $\bar{1}$ (a) and monoclinic *C*2/*c* (b) space groups, were

formed hydrothermally at 150 °C for 30 h, depending on whether $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ or $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ was added to the water, besides $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ and terpy, respectively (Scheme 1). By hydrothermal conditions iron(III) from initial building block $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ is reduced to the divalent state,¹⁷ creating 2D $[\text{Fe}^{\text{II}}_2(\text{C}_2\text{O}_4)_3]_n^{2n-}$ networks which are further connected to the complex cations $[\text{Cu}(\text{H}_2\text{O})(\text{terpy})]^{2+}$, leading to the original and unique structures of 4a and 4b.

Molecular and Crystal Structure of Compounds 1–4. *Compound $[\text{Cu}_4(\text{terpy})_4\text{Cl}_5][\text{Fe}^{\text{III}}(\text{C}_2\text{O}_4)_3] \cdot 10\text{H}_2\text{O}$ (1).* Compound 1 crystallizes in the triclinic space group $P\bar{1}$ and its crystal structure consists of a monochloro-bridged tetranuclear cation of copper(II) ions, $[\text{Cu}_4(\text{terpy})_4(\mu\text{-Cl})_3\text{Cl}_2]^{3+}$, a mononuclear oxalate anion of iron(III), $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (Figures 1 and S1), and 10 molecules of water of

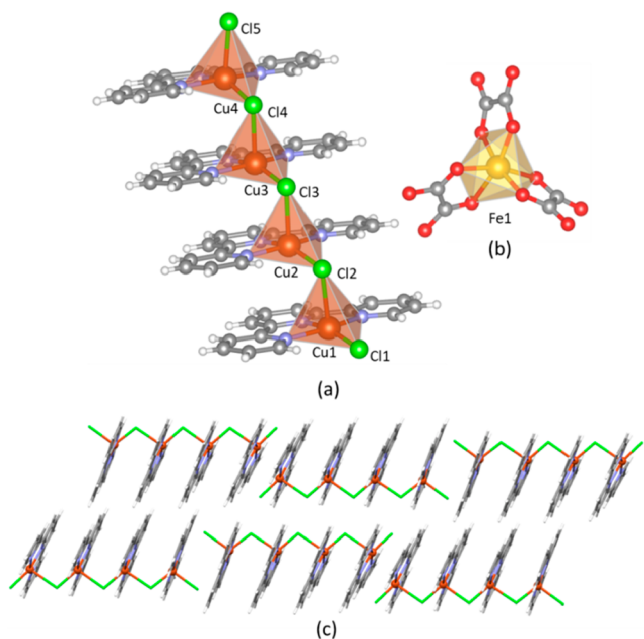


Figure 1. Monochloro-bridged tetranuclear $[\text{Cu}_4(\text{terpy})_4(\mu\text{-Cl})_3\text{Cl}_2]^{3+}$ cation (a) and the mononuclear $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ anion (b) of compound 1, showing coordination polyhedra around metal centers. The Cu and Fe atoms are depicted in orange and yellow colors, respectively. Crystal packing of the cations $[\text{Cu}_4(\text{terpy})_4(\mu\text{-Cl})_3\text{Cl}_2]^{3+}$ in the $[101]$ direction, arranged through the interactions of the aromatic systems (c).

crystallization. The complexes of copper(II) exhibiting only one chloride bridge are quite scarce, and almost all of these compounds have a 1D linear structure, with the chloride bridge placed at the equatorial-axial positions in the chain.¹⁹ The unusual zigzag asymmetric $[\text{Cu}_4(\text{terpy})_4\text{Cl}_5]^{3+}$ cation in 1 contains four copper(II) centers bridged by chloride anions, each displaying a square-pyramidal CuN_3Cl_2 coordination. The coordination sphere of atoms Cu2 and Cu3 includes three N atoms from the terpy molecule and two bridging chloride ions, while atoms Cu1 and Cu4 are coordinated, beside the three N-donor atoms from terpy ligand, by one bridging and one terminal/peripheral chloride ion (Figures 1a and S1). The distortion index (τ) was calculated for each copper(II) center using the formula $\tau = (\beta - \alpha)/60$ (α and β are the largest basal angles),²⁰ which indicates distortions from the ideal square pyramidal coordination ($\tau = 0$). The obtained values (0.033–

0.143) confirmed the square pyramidal coordination around each of the four copper centers in compound 1.

The Cu–Cl distances, in which the chloride ions are terminal, are 2.1994(11) Å (Cu1–Cl1) and 2.4589(15) Å (Cu4–Cl5); Cl5 is in apical position of the coordination geometry of Cu4, so this distance is longer, which is to be expected (Figure 1a).^{19,21} The values of the Cu–Cl distances [2.7430(15), 2.6189(14), and 2.5783(14) Å], in which bridging chloride ions are located in the apical positions of the square-pyramidal environment (Cu1–Cl2, Cu2–Cl3, and Cu3–Cl4, respectively; Figures 1a and S1) are longer than the corresponding bonds in which bridging chloride ions are located in the basal plane [Cu2–Cl2, Cu3–Cl3, and Cu4–Cl4 are 2.2265(11), 2.2444(11), and 2.2793(11) Å, respectively].²² The Cu–N distances [1.948(3)–2.033(3) Å] are similar to those found in related copper(II) complexes containing coordinated terpyridine ligand.^{23–25} Detailed geometric parameters describing the coordination polyhedra in cation and anion are given in Table S2.

The Cu...Cu distances through three chloride bridges are 3.8583(13) Å (Cu1...Cu2), 3.7605(12) Å (Cu2...Cu3), and 3.8537(12) Å (Cu3...Cu4), which are somewhat shorter than in the similar compounds with a monochloro-bridge, containing a coordinated tridentate molecule.^{26–28} The Cu–Cl–Cu bridging angles are in the range 101.01(3)–104.83(3)° (Table S2); the corresponding reported for complexes which exhibit $\text{CuY}_3(\mu\text{-X})$ cores as repeating units are 114–145°.²⁸

The iron(III) in $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ displays an approximately octahedral coordination geometry, involving six oxygen atoms from the three bidentate oxalate groups (Figures 1b and S1, Table S2). The average Fe–O distance is somewhat shorter than literature data.^{29,30}

The chloride bridging showed priority over oxalate bridging in structure 1 also due to stabilization by interactions between the aromatic rings from the coordinated terpy molecules appearing within the cation itself. Also, stacking between neighboring cations in the $[\bar{1}01]$ direction is noted, thus forming zigzag 1D chains (Figure 1c). The details of aromatic stacking interactions are given in Table S3. Furthermore, a series of intra- and intercationic C–H...Cl contacts holds together the $[\text{Cu}_4(\text{terpy})_4\text{Cl}_5]^{3+}$ units in the (110) plane. Compound 1 comprises 10 symmetry-independent uncoordinated water molecules, representing ca. 9% of the total volume. Therefore, the structure is stabilized by an extensive 3D array of O–H...O and O–H...Cl hydrogen bonds. It is certain that these water molecules connect $[\text{Cu}_4(\text{terpy})_4\text{Cl}_5]^{3+}$ and $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ ions. However, due to a large size of the asymmetric unit and presence of five transition metal atoms protons bound to water molecules could not be located in the difference Fourier maps, precluding possibility of refining them. A partial list of probable hydrogen bonds (inferred by intermolecular O...O distances of 2.6–3.0 Å) is given in Table S5.

Compound $[\text{Cu}^{\text{II}}_2(\text{H}_2\text{O})_2(\text{terpy})_2(\text{C}_2\text{O}_4)][\text{Cu}^{\text{II}}\text{Fe}^{\text{III}}(\text{CH}_3\text{OH})(\text{terpy})(\text{C}_2\text{O}_4)_2]$ (2). The structure of compound 2, crystallizing in the triclinic space group $P\bar{1}$, contains one homodinuclear oxalate-bridged $[\{\text{Cu}(\text{H}_2\text{O})(\text{terpy})\}_2(\mu\text{-C}_2\text{O}_4)]^{2+}$ cation and two heterodinuclear oxalate-bridged $[\text{Cu}(\text{CH}_3\text{OH})(\text{terpy})(\mu\text{-C}_2\text{O}_4)\text{Fe}(\text{C}_2\text{O}_4)_2]^-$ anions. There is a crystallographic inversion center in the middle of the oxalate bridge of the cation; in this way, just one-half of the cationic unit and only one of the two anionic units are symmetry-independent (Figures 2 and S2).

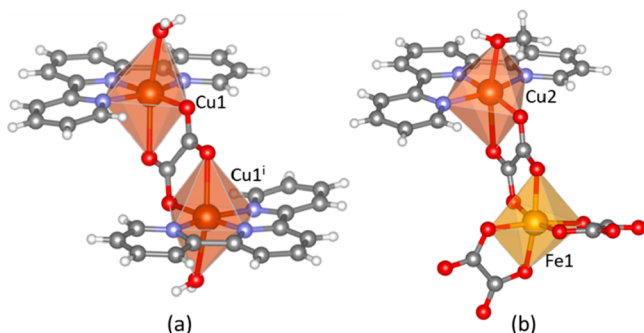


Figure 2. Homodinuclear oxalate-bridged $[\text{Cu}(\text{H}_2\text{O})(\text{terpy})(\mu\text{-C}_2\text{O}_4)\text{Cu}^i(\text{H}_2\text{O})(\text{terpy})]^{2+}$ cation [symmetry operator: (i) $1 - x, 1 - y, 2 - z$] (a) and the heterodinuclear oxalate-bridged $[\text{Cu}(\text{CH}_3\text{OH})(\text{terpy})(\mu\text{-C}_2\text{O}_4)\text{Fe}(\text{C}_2\text{O}_4)_2]^{-}$ anion (b) of compound **2**, showing coordination polyhedra around metal centers. The Cu and Fe atoms are depicted in orange and yellow colors, respectively.

Copper(II) atoms in the symmetric complex cation $[\text{Cu}(\text{H}_2\text{O})(\text{terpy})(\mu\text{-C}_2\text{O}_4)\text{Cu}^i(\text{H}_2\text{O})(\text{terpy})]^{2+}$ [symmetry operator: (i) $1 - x, 1 - y, 2 - z$] display elongated octahedral coordination, involving three N atoms from the terpy molecule [$\text{Cu1-N} = 1.934(4)\text{--}2.062(4) \text{ \AA}$] and one O atom from the bridging bis(bidentate) oxalate group [$\text{Cu1-O12} = 1.928(3) \text{ \AA}$] in the basal plane, together with other O atom of bridging group [$\text{Cu1-O11} = 2.357(3) \text{ \AA}$] and one O atom of coordinated water [$\text{Cu1-O13} = 2.375(4) \text{ \AA}$] in the apical positions (Figures 2a and S2). A similar elongated octahedral geometry copper(II) atom exists in the heterometallic $[\text{Cu}(\text{CH}_3\text{OH})(\text{terpy})(\mu\text{-C}_2\text{O}_4)\text{Fe}(\text{C}_2\text{O}_4)_2]^{-}$ anion; it is surrounded by three N atoms from a terpy ligand and one O atom of the oxalate bridge, having four short equatorial bonds [$\text{Cu2-N} = 1.928(4)\text{--}2.026(4) \text{ \AA}$; $\text{Cu2-O15} = 1.974(3) \text{ \AA}$] and by two O atoms, one from oxalate bridge [$\text{Cu1-O14} = 2.424(4) \text{ \AA}$] and the other from coordinated methanol solvate [$\text{Cu1-O16} = 2.348(4) \text{ \AA}$] with two longer axial bonds (Figure 2b). The iron(III) ion is coordinated by six O atoms; four from two bidentate oxalate groups [$\text{Fe1-O} = 1.988(4)\text{--}2.000(3) \text{ \AA}$] and two from oxalate bridge [Fe1-O are $2.029(3)$ and $2.081(3) \text{ \AA}$].^{31,32} Selected distances and angles in the complex cation and anion of compound **2** are given in Table S4.

The $\text{Cu1}\cdots\text{Cu1}^i$ [symmetry operator: (i) $1 - x, 1 - y, 2 - z$] and $\text{Cu2}\cdots\text{Fe1}$ distances across the oxalate bridges are $5.4998(12)$ and $5.5019(9) \text{ \AA}$, respectively. Each coordinated water from cation is involved in hydrogen bonds with uncoordinated oxygen atoms of oxalate groups of two different symmetric neighboring anionic units, forming a ladderlike hydrogen bonding motif along the a -axis (Figure S3a). Furthermore, the hydroxyl group of the coordinated methanol molecule and oxygen atoms (one uncoordinated and one coordinated) from oxalate group from two anionic units participate in hydrogen bonding, resulting in 1D arrangement parallel to the b -axis (Figure S3b), thus forming an overall 2D hydrogen bonding architecture. The details describing the hydrogen bonding geometry are given in Table S5.

The crystal structure of **2** is stabilized by stacking interactions between the aromatic rings in the terpy molecules of cations and between those of anions (Table S3). The rings of the neighboring $[\{\text{Cu}(\text{H}_2\text{O})(\text{terpy})\}_2(\mu\text{-C}_2\text{O}_4)]^{2+}$ species are stacked along $[010]$ direction forming alternating, sandwichlike π -stacks (Figure S4a), while those of the

neighboring $[\text{Cu}(\text{CH}_3\text{OH})(\text{terpy})(\mu\text{-C}_2\text{O}_4)\text{Fe}(\text{C}_2\text{O}_4)]^{-}$ anions generate 1D motif along the $[110]$ direction (Figure S4b).

Compound $\{[\text{Cu}^{\text{II}}_2\text{Fe}^{\text{III}}(\text{H}_2\text{O})(\text{terpy})_2(\text{C}_2\text{O}_4)_{7/2}]\cdot 6\text{H}_2\text{O}\}_n$ (**3**). Structural analysis has showed that in compound **3**, crystallizing in the triclinic space group $P\bar{1}$, the coordinated oxygen atom from the oxalate group of the $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ anion is connected to the copper(II) atom of the $[\text{Cu}(\text{terpy})(\text{H}_2\text{O})]^{2+}$ unit, while the terminal one of the other oxalate group to the copper(II) atom from the symmetric unit $[\text{Cu}(\text{terpy})(\mu\text{-C}_2\text{O}_4)\text{Cu}^i(\text{terpy})]^{2+}$ [symmetry operator: (ii) $2 - x, 1 - y, 2 - z$]. There is a crystallographic inversion center in the middle of the oxalate bridge of this cation. Oxygen atoms from the oxalate bridge are coordinated to the copper(II) atoms of the $[\text{Cu}(\text{terpy})(\text{H}_2\text{O})]^{2+}$ units, thus forming a 1D ladderlike $[\text{Cu}^{\text{II}}\text{Fe}^{\text{III}}]$ coordination polymer along the a -axis (Figures 3 and S5).

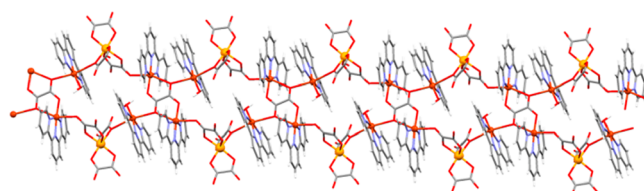


Figure 3. Ladderlike 1D chain motif in compound $\{[\text{Cu}^{\text{II}}_2\text{Fe}^{\text{III}}(\text{H}_2\text{O})(\text{terpy})_2(\text{C}_2\text{O}_4)_{7/2}]\cdot 6\text{H}_2\text{O}\}_n$ (**3**) along the a axis.

The Cu1 atom, besides the two O atoms from two oxalate groups of the $[\text{Cu}(\text{terpy})(\mu\text{-C}_2\text{O}_4)\text{Cu}^i(\text{terpy})]^{2+}$ and $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ units in apical positions [$\text{Cu1-O14} = 2.522(2) \text{ \AA}$ and $\text{Cu1-O6}^i = 2.7258(18) \text{ \AA}$, respectively; symmetry operator: (i) $1 - x, 1 - y, 1 - z$], is coordinated by three N atoms from the terpy ligand [$\text{Cu1-N} = 1.916(2)\text{--}2.040(3) \text{ \AA}$] and O atom of water molecule [$\text{Cu1-O15} = 1.928(2) \text{ \AA}$] (Figures 3 and 4; Table S6). The observed axial Cu1–O distances are significantly longer than the typical Cu–O covalent bonds ($1.98 \text{ \AA}$),¹ but they are considerably shorter than the sum of the van der Waals radii ($2.92 \text{ \AA}$).³³

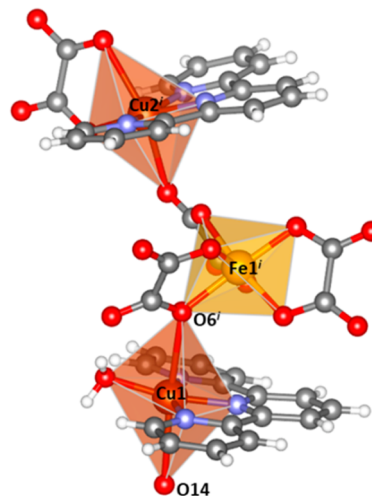


Figure 4. Coordination polyhedra around metal centers (Fe1, Cu1 and Cu2) in compound $\{[\text{Cu}^{\text{II}}_2\text{Fe}^{\text{III}}(\text{H}_2\text{O})(\text{terpy})_2(\text{C}_2\text{O}_4)_{7/2}]\cdot 6\text{H}_2\text{O}\}_n$ (**3**). The Cu and Fe atoms are depicted in orange and yellow colors, respectively. Symmetry operator: (i) $1 - x, 1 - y, 1 - z$.

As in compound **2** described above, the oxalate release of $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and its delivery allow us to obtain symmetrical oxalate-bridged cations $[\text{Cu}(\text{terpy})(\mu\text{-C}_2\text{O}_4)\text{Cu}^{\text{II}}(\text{terpy})]^{2+}$, in which copper(II) ion displays elongated octahedral coordination, involving three N atoms from the terpy molecule $[\text{Cu2-N} = 1.928(2)\text{--}2.049(2) \text{ \AA}]$ and one O atom from the bridging bis(bidentate) oxalate group in the basal plane $[\text{Cu2-O13} = 1.9326(18) \text{ \AA}]$, together with another O atom from the same oxalate bridge $[\text{Cu2-O14}^{\text{ii}} = 2.4229(19) \text{ \AA}]$ and one from terminal oxalate group of the $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ anion $[\text{Cu2-O12} = 2.398(2) \text{ \AA}]$ in apical positions (Figures 3, 4, and S5). The iron(III) ion from the bridging $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ building block displays an approximately octahedral coordination geometry, involving six O atoms from three oxalate groups. The values of the Fe–O bond lengths are in the relatively good agreement with literature data.³⁴ Selected bond distances and angles for **3** are listed in Table S6.

Uncommonly, three different bridging modes of the oxalate group are observed in compound **3**: bidentate-monodentate $[\mu\text{-}1,1,2]$ and bidentate-monodentate $[\mu\text{-}1,2,3]$ between copper(II) and iron(III) centers and bis(bidentate)–bis(monodentate) $[\mu_4\text{-}1,1,2,3,3,4]$ between four copper centers (Scheme S1a).¹⁶

1D coordination polymer **3** is extended in the direction $[100]$ by six symmetry-independent uncoordinated water molecules arranged into discrete centrosymmetric clusters $(\text{H}_2\text{O})_{12}$ (Figures S6 and S7); each of these clusters forms hydrogen bonds with six neighboring 1D chains of **3**, forming a 3D array (Table S5 and Figure S7).

Compound $[\text{Cu}^{\text{II}}\text{Fe}^{\text{II}}_2(\text{H}_2\text{O})(\text{terpy})(\text{C}_2\text{O}_4)_3]_n$ (**4**). Two polymorphs of the coordination polymer $[\text{CuFe}_2(\text{H}_2\text{O})(\text{terpy})(\text{C}_2\text{O}_4)_3]_n$ (**4**), one crystallizing in the triclinic $P\bar{1}$ space group (**4a**) and other in the monoclinic $C2/c$ (**4b**), comprise a 3D structural arrangement. Polymers contain homometallic 2D oxalate-bridged honeycomb anionic layers with (6^3) or hcb topology, $[\text{Fe}^{\text{II}}_2(\text{C}_2\text{O}_4)_3]_n^{2n-}$, mutually linked by complex cations of copper(II) ions, $[\text{Cu}(\text{H}_2\text{O})(\text{terpy})]^{2+}$, through oxygen atoms from oxalate bridges (Figure 5; Scheme S1b). Thus, the overall 3D structural network of a novel topology ($12^3\cdot 2\cdot 12^5\cdot 12\cdot 3$) is formed in both compounds (Figure S8).³⁵ The structure of **4a** contains one symmetric $[\text{Cu}(\text{H}_2\text{O})(\text{terpy})]^{2+}$ cation and one symmetric $[\text{Fe}_2(\text{C}_2\text{O}_4)_3]^{2-}$ anion mutually bridged, while in compound **4b**, just one-half of the cationic unit and one-half of the anionic unit are symmetry-independent (Figures 6 and S9).

The Fe^{II} atoms in both polymorphs are coordinated by six O atoms from three bridging oxalate ligands, displaying approximately octahedral coordination geometry (Figure 6); the Fe–O distances fall into the range of the lengths for only one known oxalate-bridged 2D honeycomb anion of Fe^{II} containing the 5-oxo-1,4,7-triazabicyclo[4.3.0]non-6-en-7-yl ammonium ion as a templated cation.³⁶ The bis(bidentate) oxalate ligands repeatedly bridge adjacent Fe^{II} ions in all three dimensions, which finally leads to a polymeric net parallel to the (011) plane in **4a** and to (100) plane in **4b**. In Table S7, selected bond lengths and angles for polymers **4a** and **4b** are presented. Each honeycomb of the $[\text{Fe}^{\text{II}}_2(\text{C}_2\text{O}_4)_3]_n^{2n-}$ layer in **4a** and **4b** contains 6 oxalate bridges; two diagonal O atoms from each bridge in the $[100]$ direction for **4a** and the $[101]$ direction for **4b** are coordinated to two copper(II) ions (Figures 5a and 5c). Thus, two different bridging modes of the oxalate group are observed in compounds **4a** and **4b**: bis(bidentate), $[\mu\text{-}1,2,3,4]$, between two iron(II) ions and

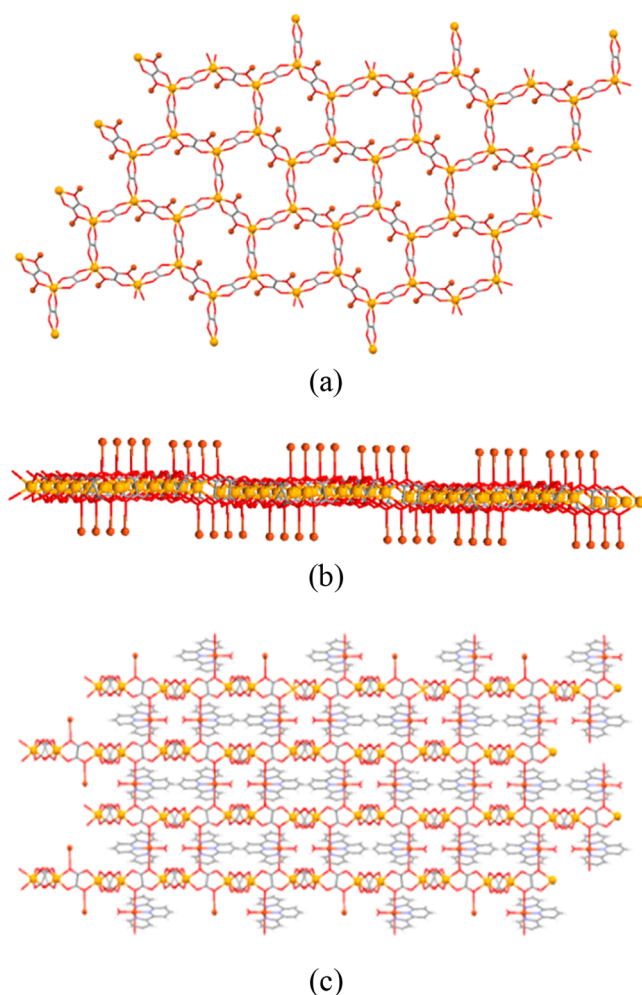


Figure 5. 2D honeycomb anionic network $[\text{Fe}^{\text{II}}_2(\text{C}_2\text{O}_4)_3]_n^{2n-}$ of compound **4a** connected to the copper ions in the (011) plane (a) and in the [011] direction (b). Water and terpyridine molecules are omitted for clarity. Coordination of the oxygen atoms from oxalate bridges of the $[\text{Fe}^{\text{II}}_2(\text{C}_2\text{O}_4)_3]_n^{2n-}$ network to the copper(II) ions from the $[\text{Cu}(\text{H}_2\text{O})(\text{terpy})]^{2+}$ cations in the $[100]$ direction (c).

bis(bidentate)–bis(monodentate) $[\mu_4\text{-}1,1,2,3,3,4]$ between two copper(II) and two iron(II) centers (Scheme S1b).

The copper(II) atom from the complex cation displays elongated octahedral coordination, involving three N atoms from the terpy molecule $[1.908(7)\text{--}2.028(7) \text{ \AA}]$ for **4a** and $1.909(3)\text{--}2.012(2)$ for **4b**; Table S7] and one O atom from coordinated water molecule $[1.936(6)$ and $1.931(3) \text{ \AA}]$ for **4a** and **4b**, respectively] in the basal plane, together with O atoms from the bridging oxalate groups from the neighboring $[\text{Fe}^{\text{II}}_2(\text{C}_2\text{O}_4)_3]_n^{2n-}$ layers in the apical positions (Figure 6). The observed axial Cu–O distances $[\text{Cu1-O9} = 2.618(7) \text{ \AA}]$ and $\text{Cu1-O11}^i = 2.619(6) \text{ \AA}]$; symmetry operator: $(i) 1 - x, 1 - y, 1 - z$ for **4a** and $\text{Cu1-O6} = \text{Cu1-O6}^i = 2.6233(16) \text{ \AA}]$; symmetry operator: $(i) 1 - x, y, 3/2 - z$ for **4b**; Table S7] are slightly elongated, but it are still within a typical range ($\sim 2.32\text{--}2.80 \text{ \AA}$) indicating the semicoordination character of the Cu–O bonds.¹

Up to now, all known 2D honeycomb anionic layers in the compounds $\{(\text{A})_2[\text{M}^{\text{II}}_2(\text{C}_2\text{O}_4)_3]\}_n$ or $\{\text{A}[\text{M}^{\text{II}}\text{M}^{\text{III}}(\text{C}_2\text{O}_4)_3]\}_n$ were only interleaved by the templating cations,¹ except in compounds $\{[\text{Cu}(\text{en})_2][\text{Mn}_2(\text{C}_2\text{O}_4)_3]\}_n$, $\{[\text{Cu}(\text{en})_2][\text{Co}_2(\text{C}_2\text{O}_4)_3]\cdot 3\text{H}_2\text{O}\}_n$ and $\{[\text{Cu}(\text{Me}_2\text{en})_2][\text{Mn}_2(\text{C}_2\text{O}_4)_3]\}_n$.

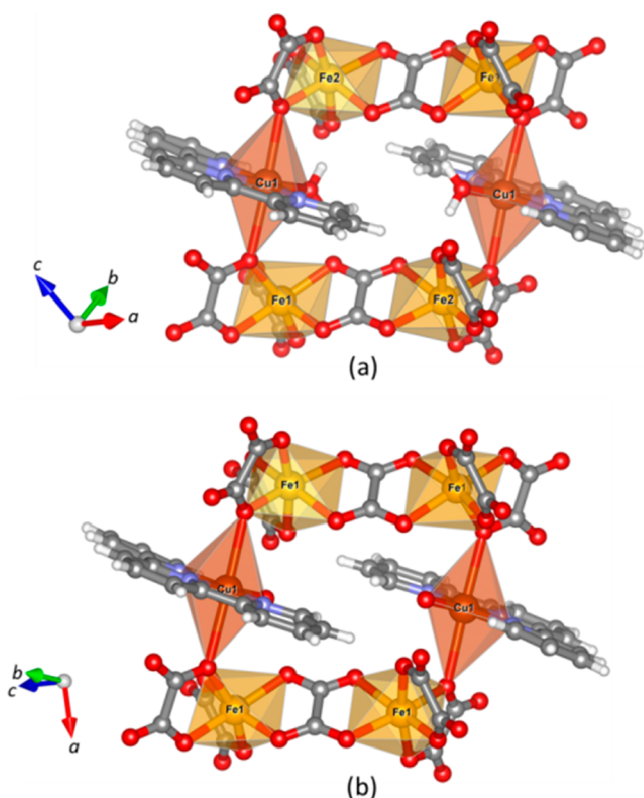


Figure 6. Coordination polyhedra around metal centers in the polymorph **4a** (a) and **4b** (b). The Cu and Fe atoms are depicted in orange and yellow colors, respectively.

$2\text{H}_2\text{O}\}_n$ (en = ethylenediamine; Me₂en = *N,N*-dimethylethylenediamine), in which the cationic units are linked with the layers.^{37,38}

The average Fe...Fe separations across the oxalate bridges are 5.534 and 5.5486 Å in **4a** and **4b**, respectively. Additionally, distance between Cu^{II} and Fe^{II} metal centers bridged by oxygen atoms from oxalate bridges are 4.172(3) Å [Cu1...Fe1] and 4.174(4) Å [Cu1...Fe2ⁱ; symmetry operator (*i*): 1 - *x*, 1 - *y*, 1 - *z*] in polymorph **4a** and 4.1823(4) Å [Cu1...Fe1ⁱ; symmetry operator (*i*) -*x* + 1/2, -*y* + 1/2, -*z* + 1] in polymorph **4b**. The distance between two honeycomb anionic layers in **4a** is 7.544 Å and in **4b** 7.553 Å, and these interlayer separations are significantly shorter than in the similar compounds.^{37,38}

The thermal behavior of compounds **1**, **2**, and **4a** has been studied by simultaneous thermogravimetric analysis (TG) and differential thermal analysis (DTA) performed on a crystalline sample in a stream of synthetic air, up to 1000 °C. The corresponding curves are presented in Figure S10 and data in Table S8. All compounds start to decompose almost immediately after the beginning of heating when lose water and/or methanol molecules. In **1**, after dehydration, there is a main loss of mass characterized by strong exothermic effects, ascribed to the release of chloride ions and terpyridine molecules and to the oxalate ligands between 450 and 600 °C. Strong exothermic effects in compound **2** can be attributed to the release of terpyridine and oxalate ligands (200–600 °C). The TGA curve of **4a** shows single-step decomposition ending around 600 °C. This step, followed by strong exothermic DTA maximum, could be associated with the removal of water and terpyridine molecules, together with the elimination of the

oxalate groups. Crystalline product obtained by pyrolysis of this 3D coordination polymer up to 1000 °C was explored by powder X-ray diffraction at room temperature (Figure S11). Unfortunately, not only desired spinel CuFe₂O₄ oxide (81.6%), but also CuO (13.3%) and Fe₂O₃ (5.1%) phases were formed.

Infrared Study of Compounds 1–4. The IR spectra of compounds **1–4** show characteristic absorption bands of the oxalate groups and coordinated *N*-donor ligand. All absorption bands observed in the IR are described in the Experimental Section of the Supporting Information. The absorption bands corresponding to the stretching vibrations of the coordinated [bis(bidentate) and bis(bidentate)–bis(monodentate)] and uncoordinated [bidentate and bidentate–monodentate] CO groups of the oxalate ligands^{16,34} for all compounds are summarized in Table S9. The terminal and bridging Cu–Cl stretching vibrations in compound **1** appear at 324 and 308 cm⁻¹, respectively.³⁹

Electrical Properties of Compounds **1**, **2**, **4a**, and **4b**.

The complex impedance plots of compounds **1** and **4b** at 30 and 40 °C, respectively, are shown in Figure 7. Compound **1**

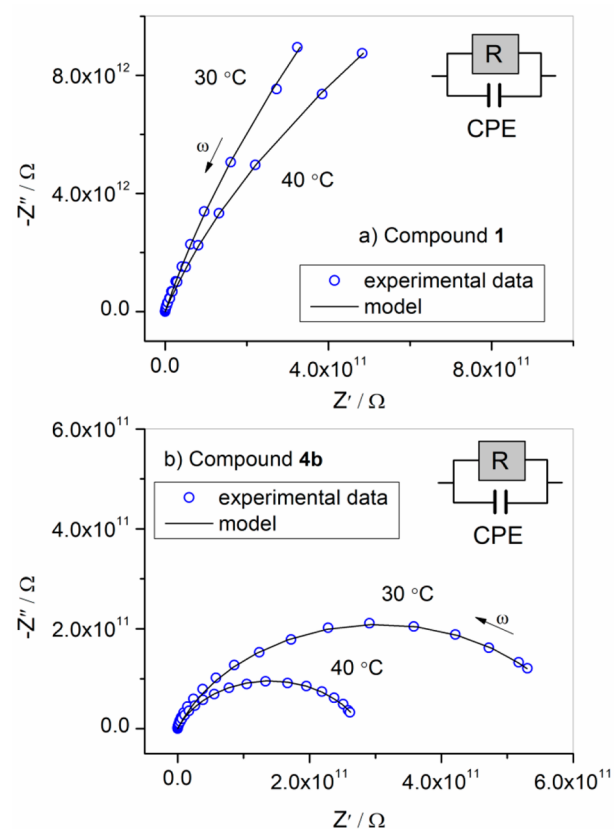


Figure 7. Complex impedance plot and corresponding equivalent circuit for compounds **1** (a) and **4b** (b) at 30 and 40 °C.

exhibits arcs at high values of impedance indicating low electrical conductivity (Figure 7a), whereas **4b** shows well-defined impedance semicircles at much lower values of impedance (Figure 7b). Similar result was obtained for compounds **2** and **4a**, respectively (Figure S12). The obtained impedance data can be approximated well by the equivalent electrical circuit consisting of a parallel combination of resistor and constant phase element (CPE). From the values of electrical resistance (*R*) obtained from equivalent circuit modeling and electrode dimensions (*A* is the electrode area

and d is the sample thickness) the values of direct current (DC) conductivity for all compounds were calculated according to relation: $\sigma_{\text{DC}} = d/(A \cdot R)$. It is worth noting that equivalent circuit modeling was especially useful for assessing the σ_{DC} of compounds **1** and **2** for which impedance spectra do not exhibit semicircles; hence, the low-frequency intercept on the x -axis corresponding to DC resistance could not be attained (see Figure 7a).

The temperature dependences of DC conductivity, σ_{DC} , of compounds **1**, **2**, **4a**, and **4b** are shown in an Arrhenius type plot of $\log(\sigma_{\text{DC}})$ versus $1000/T$ (Figure 8). Compounds **1** and

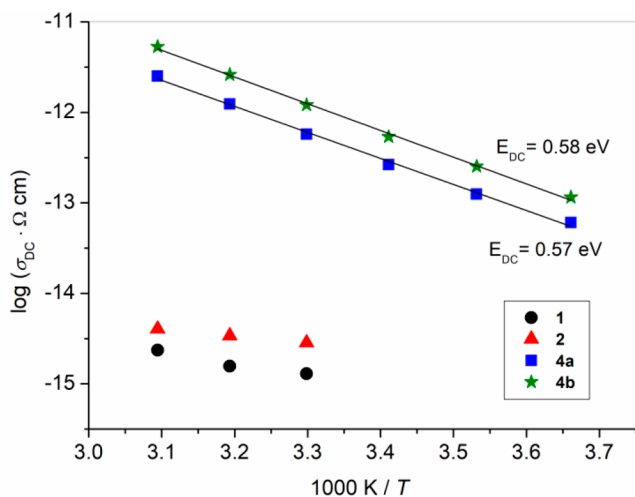


Figure 8. Electrical conductivity as a function of reciprocal temperature for **1**, **2**, **4a**, and **4b**. Solid lines represent least-squares linear fit to experimental data.

2 exhibit insulating behavior with similarly very low values of conductivity [$\approx 10^{-15} (\Omega \text{ cm})^{-1}$] over temperatures from 30 to 50 °C. The values of σ_{DC} at lower temperatures (< 30 °C) for these two compounds could not be determined due to too high impedance. In contrast, polymers **4a** and **4b** show more than 2 orders of magnitude higher electrical conductivity, reaching the values of $2.6 \times 10^{-13} (\Omega \text{ cm})^{-1}$ and $5.4 \times 10^{-13} (\Omega \text{ cm})^{-1}$ at 20 °C, respectively. As can be seen from Figure 8, the conductivity of **4a** and **4b** is thermally activated and from the slope the activation energy for DC conductivity, E_{DC} , is calculated according to the Arrhenius relation, $\sigma_{\text{DC}} = \sigma_0 \exp(-E_{\text{DC}}/k_B T)$. The values of E_{DC} for both **4a** and **4b** are found to be very similar, 0.57 and 0.58 eV, respectively. The difference in electrical conductivity between two sets of compounds, namely, **1/2** and **4a/4b** is directly related to their structural diversity.

The well-defined crystalline structures of coordination polymers can enable viewing of the exact proton-conducting pathways, giving deeper insight into their electrical properties. Proton conductivity requires proton carriers such as H_3O^+ , NH_4^+ , or H^+ belonging to acid groups or to networks of hydrogen bonds. The simplest method to obtain conductive materials is to introduce proton carriers directly into polymers as counterions. The strategy developed for the oxalate-based compounds relies on the introduction of acidic or hydrophilic residues into the cationic component in order to construct proton-conductive pathways between the layers.^{40–43}

Indeed, compounds **4a** and **4b** exhibit a strong increase of the electrical conductivity with the increase in the relative

humidity (see Figure 9), confirming the existence of proton conductivity most probably due to hydrogen-bonding between

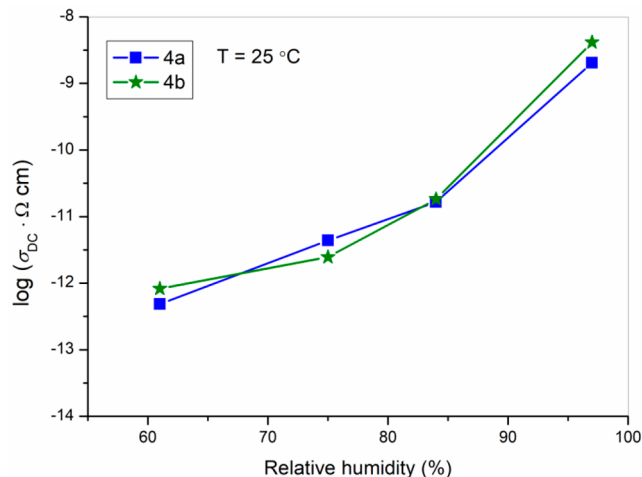


Figure 9. Conductivity as a function of relative humidity (%) for **4a** and **4b** at 25 °C.

hydrogen atoms as a proton source from the water molecule coordinated to copper(II) ions, $[\text{Cu}(\text{H}_2\text{O})(\text{terpy})]^{2+}$, and oxalate oxygen atoms from two neighboring honeycomb layers $[\text{Fe}^{II}_2(\text{C}_2\text{O}_4)_3]^{2n-}$ (Figure S13 and Table S5).

Measured conductivities can be also correlated to the electronic structure obtained from DFT calculations. Figure S14 shows density of states of compounds **1**, **2**, **3**, and **4b**. It can be seen that DFT predicts electronic band gaps of 1.8 eV for **1**, 0.6 eV for **2**, 0.8 eV for **3**, and 0.1 eV for **4b**. Note that DFT at this level is known to underestimate electronic gaps. Nevertheless, the small electronic band gap found for **4b** is in line with a significantly higher conductivity of this compound.

Magnetic Properties of Compounds **1**, **2**, **4a**, and **4b**.

The $\chi T(T)$ dependences for **1** is shown in Figure 10. By increasing the temperature from 2 K as the lowest, χT is rapidly increasing to its maximum value at about 14 K, after

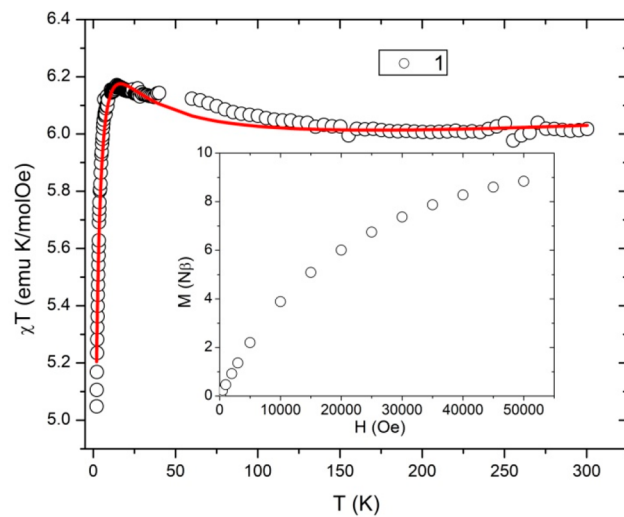


Figure 10. Temperature dependence of product χT for compound **1**. The red solid line is model curve. Inset: Measured $M(H)$ dependence at $T = 2$ K for **1**.

which it slightly falls before becoming the final straight horizontal line at high temperatures.

The $M(H)$ curve measured at 2 K is not fully saturated even in the largest measured field of 50 kOe (inset in Figure 10). Value of the magnetic moment at 50 kOe is around $8.85N\mu_B$. The expected value of the saturated moment for four Cu^{2+} centers with spin $S = 1/2$ from $[\text{Cu}_4(\text{terpy})_4\text{Cl}_5]^{3+}$ cation and one Fe^{3+} with spin $S = 5/2$ from $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ anion is $(4g_{\text{Cu}} \cdot 1/2 + g_{\text{Fe}} \cdot 5/2)N\mu_B$, which is slightly higher than the measured one.

Magnetic interactions of copper(II) centers mediated by one^{19,21,26,27,44–48} or two Cl[−] bridges^{22,24,44,47,49,50} have been studied. The most promising structural feature that determines the character and strength of the magnetic interaction is the ratio between the angle at the Cu^{II}–Cl–Cu^{II} bridge (θ) and longer Cu–Cl distance (R). There is a simple correlation between the exchange parameter J and this ratio θ/R . Overall, ferromagnetic behavior can be expected for values of the ratio θ/R lower than approximately 40 and higher than 58 (in units of degrees/Å), while antiferromagnetic character appears when this quotient θ/R is between these two values.⁴⁸ The values of the ratios θ/R found in complex salt 1 (Table S2) are 37.95 (Cu1/Cu2), 38.56 (Cu2/Cu3), and 40.65 (Cu3/Cu4), announcing the existence of both ferro- and antiferromagnetic interactions between copper(II) ions through chloride bridges. These values are close to the border value between the ferro- and antiferromagnetic characters of the interaction. Therefore, to reproduce the magnetic behavior of 1 all three different Cu^{II}–Cl–Cu^{II} interactions (Cu1...Cu2, Cu2...Cu3, and Cu3...Cu4) have been taken in the model. The magnetic behavior of 1 is modeled by the Hamiltonian:

$$H = -J_1 S_1 \cdot S_2 - J_2 S_2 \cdot S_3 - J_3 S_3 \cdot S_4 + \sum_{i=1}^5 g_i \mu_B H \cdot S_i \quad (1)$$

where S_1 , S_2 , S_3 , and S_4 are spins of Cu^{2+} and S_5 is the Fe^{3+} spin, and the other symbols have their usual meaning. The zero-field splitting contribution of Fe^{3+} is expected to be negligible. The best-fit modeled curve reproduces the measured data well (Figure 10); the best fitting was obtained for the following parameters: $g_{\text{Cu}} = 2.075$, $g_{\text{Fe}} = 1.98$, $J_1 = 23.98 \text{ cm}^{-1}$, $J_2 = 16.43 \text{ cm}^{-1}$, and $J_3 = -5.01 \text{ cm}^{-1}$. Obtained results suggest dominant ferromagnetic interactions over antiferromagnetic interaction across the Cu^{2+} tetramer. The values of the interactions J_1 and J_2 indicate so far the strongest ferromagnetic interaction in (μ -chloro)copper(II) compounds.⁴⁸ Numerical calculation of the Hamiltonian model treats different Cu magnetic centers as indistinguishable, but according to the proposed magneto-structural correlation, these interactions nevertheless would lead to the spin $S = 1$ of the ground state determining therefore its magnetic character. Obtained values of the magnetic interactions J are in agreement with those obtained from the DFT calculations, which also confirm competing ferromagnetic and antiferromagnetic interactions resulting with a $S = 1$ ground-state spin.

The $\chi T(T)$ dependence for compound $[\text{Cu}^{\text{II}}_2(\text{H}_2\text{O})_2(\text{terpy})_2(\text{C}_2\text{O}_4)] [\text{Cu}^{\text{II}}\text{Fe}^{\text{III}}(\text{CH}_3\text{OH})(\text{terpy})-(\text{C}_2\text{O}_4)_3]_2$ (2) is shown in Figure 11. By decreasing temperature, $\chi T(T)$ is falling in the whole temperature interval. The value of product $\chi T(T)$ at room temperature is $10.0 \text{ emu K mol}^{-1} \text{ Oe}^{-1}$. The expected moment coming from four Cu^{2+} ions (two from cation and two from two anions) with spin $S = 1/2$ and $g_{\text{Cu}} = 2.1$ and from two Fe^{3+} (from two

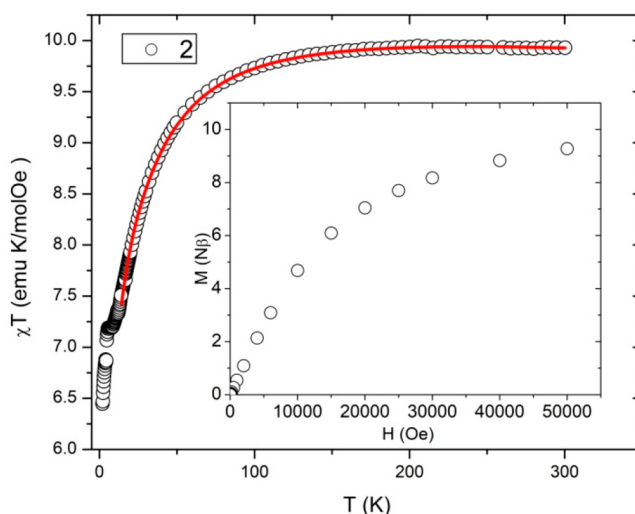


Figure 11. Temperature dependence of product χT for compound 2. The red solid line is model curve. Inset: Measured $M(H)$ dependence at $T = 2 \text{ K}$ for 2.

anions) with spin $S = 5/2$ and $g_{\text{Fe}} = 2.0$ is $10.4 \text{ emu K mol}^{-1} \text{ Oe}^{-1}$, which is a slightly higher value compared to the measured one. The $M(H)$ curve measured at 2 K is shown in the inset of Figure 11; this curve is not saturated even in the highest measured fields of 50 kOe. The value of the moment in 50 kOe is $9.23N\mu_B$, which is notably less than the value corresponding to that of saturation of four Cu^{2+} and two Fe^{3+} magnetic centers ($14.2N\mu_B$), signaling strong interaction between these ions.

Magnetic behavior could be explained by the Hamiltonian model:

$$H = -J_1 S_1 \cdot S_2 - J_2 (S_3 \cdot S_4 + S_5 \cdot S_6) + \sum_{i=1}^6 g_i \mu_B H \cdot S_i \quad (2)$$

where S_1 , S_2 , S_3 , and S_5 are the spins of Cu^{2+} and S_4 and S_6 are the spins of Fe^{3+} . The best-fit curve (shown in Figure 11) reproduces the measured data very well. Data below 14 K were excluded from fitting due to the thermal instability of the instrument which made these measured points unreliable. The best fitting was obtained for the following parameters: $g_{\text{Cu}} = 2.16$, $g_{\text{Fe}} = 1.99$, $J_1 = -72.50 \text{ cm}^{-1}$, and $J_2 = -4.62 \text{ cm}^{-1}$. The interaction between Cu^{2+} ions across oxalate group in cation $[\text{Cu}(\text{H}_2\text{O})(\text{terpy})(\mu\text{-C}_2\text{O}_4)(\text{terpy})(\text{H}_2\text{O})\text{Cu}]^{2+}$ is antiferromagnetic (J_1) as well as those between Cu^{2+} and Fe^{3+} centers (J_2) in two anions $[\text{Cu}(\text{CH}_3\text{OH})(\text{terpy})(\mu\text{-C}_2\text{O}_4)\text{Fe}-(\text{C}_2\text{O}_4)_2]^-$. A very strong interaction of order of almost 100 K between copper(II) ions mediated by the oxalate bridge is observed in present cation. This very strong interaction is significant compared to the thermal energy even at room temperature, which results in a measured nonsaturated χT value at room temperature. Such strong antiferromagnetic interactions are also confirmed from DFT calculations. However, DFT calculations suggest a somewhat stronger J_2 interaction, but with this value it was not possible to reproduce the measured temperature dependence of magnetization. It is well-known that the used DFT method usually overestimate the exchange energies.

Interestingly, search of the CSD¹ does not find any compound in which Cu^{2+} and Fe^{3+} ions are bridged by bis(bidentate) oxalate group. Also, only one compound

containing $[\{\text{Cu}(\text{H}_2\text{O})(\text{terpy})\}_2(\mu\text{-C}_2\text{O}_4)]^{2+}$ unit was found, but no magnetic investigation was done.⁵¹ Therefore, the obtained value of the magnetic interaction is a pioneer one, and in order to set magneto-structural correlations, new compounds are desirable.

In 3D coordination polymers **4a** and **4b**, both with the formula $[\text{CuFe}_2(\text{H}_2\text{O})(\text{terpy})(\text{C}_2\text{O}_4)_3]_n$, magnetic interactions are conveyed through the network and produce long-range order. Magnetic order is studied and confirmed by the more detailed magnetization measurements. The $\chi(T)$ dependences (inset of Figure 12 for **4a** and Figure S15 for **4a** and **4b**) were

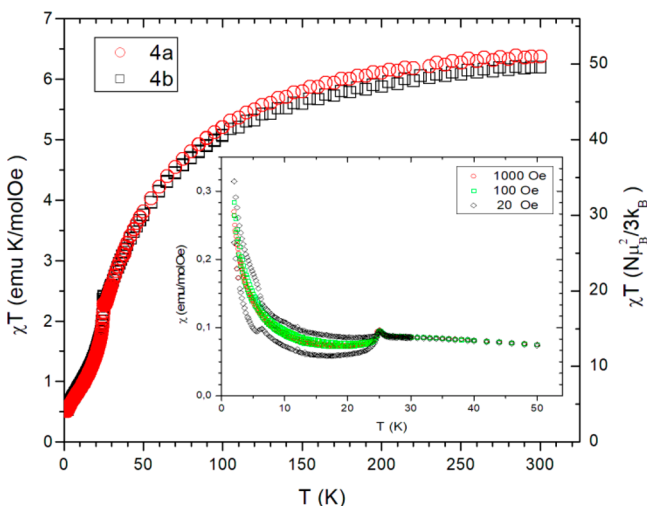


Figure 12. Temperature dependence of product χT for compounds **4a** and **4b**. Inset: $\chi(T)$ around transition temperature for **4a** in different fields, where lower branches are ZFC parts and upper branches FC parts.

measured in different magnetic fields in order to identify magnetic transition. Peaks at the $\chi(T)$ dependence show that both samples undergo antiferromagnetic transition around 25 K (inset of Figure 12). The same behavior of susceptibility is obtained at different values of external magnetic fields ranging from 20 to 1000 Oe, for both samples. Below the magnetic transition at 25 K, an easily observable splitting between the zero-field-cooled (ZFC, lower branch) and field-cooled (FC, upper branch) curves appears, which is very small for moderate fields (around 100 Oe) and unobservable for high fields (above 1000 Oe) but considerable for small fields (below 20 Oe). This splitting shows that the ordered phase has some thermal irreversibility which might be connected with weak ferromagnetism and/or canted spins, as was already observed in some 3D coordination polymers.^{14–16} The magnetic phase transition at 28 K was already observed in a previously reported polymer in which 2D honeycomb $[\text{Fe}^{\text{II}}_2(\text{C}_2\text{O}_4)_3]_n^{2n-}$ layers are templated by organic cations.³⁶

Plotting the $\chi T(T)$ dependences (Figure 12) points to relatively strong antiferromagnetic interactions in the systems. Temperature dependence of reciprocal susceptibility in the paramagnetic phase, $\chi^{-1}(T)$, for both **4a** and **4b** has linear behavior as predicted by the Curie–Weiss law $\chi = C/(T - \theta)$ (Figure S16). Fitting the data for both samples, the following Curie constants C and Weiss temperatures θ have been obtained: $C = 8.0(2) \text{ emu K mol}^{-1} \text{ Oe}^{-1}$, $\theta = -52(2) \text{ K}$ for **4a** and $8.2(3) \text{ emu K mol}^{-1} \text{ Oe}^{-1}$, $\theta = -53(3) \text{ K}$ for **4b**. These Weiss constants confirm that the antiferromagnetic inter-

actions dominate in the systems. Their values are in agreement with the exchange couplings obtained from the DFT calculations from which it could be expected that the Weiss parameter is negative and of the order $\sim 100 \text{ K}$. Only the dominant interactions between Fe^{2+} spins in the honeycomb lattice have been taken into account as those who generate the mean field in the magnetic lattice. The obtained values of C have been compared with the expected values for non-interacting spins. Using the value of $S_{\text{Fe}} = 2$ and $S_{\text{Cu}} = 1/2$ and published g -factors which are for Cu^{2+} in an elongated octahedral environment between 2 and 2.1⁵² and for Fe^{2+} have an octahedral geometry, this value can be anywhere between 1.975 and 2.245,⁵³ for the Curie constant it follows $8.1(2) \text{ emu K mol}^{-1} \text{ Oe}^{-1}$. We saw good agreement of this value with the experimentally determined points to the paramagnetic state of noninteracting spins in **4a** and **4b** at high temperatures.

At low temperature, exchange interactions between iron(II) ions through oxalate bridges in the honeycomb lattice $[\text{Fe}_2(\text{C}_2\text{O}_4)_3]_n^{2n-}$ come into account and lead to the magnetic order. The magnetic interaction between Fe^{2+} and Cu^{2+} magnetic centers is expected to be small due to relatively large distances between Cu^{2+} ions and O atoms from oxalate bridges of two neighboring layers. However, the Weiss parameters of -52 K (**4a**) and -53 K (**4b**) point out the dominant magnetic interactions which are antiferromagnetic, and their effective values are of considerable strength, although being a combination of intralayer and much weaker interlayer interactions. This is confirmed with DFT calculations, which give the values of intralayer (Fe–ox–Fe) interactions between 20–50 K with antiferromagnetic sign. The values for interlayer interaction transferred through $\text{O}_{\text{ox}}\text{--Cu--O}_{\text{ox}}$ bridges (see the section “Molecular and Crystal Structure of Compounds 1–4”), and some types of interlayer Fe...Fe interactions are much smaller, ~ 1 and $\sim 0.1 \text{ K}$ respectively.

Another useful comparison for resolving the character of the dominant magnetic interaction between Fe^{2+} centers is shown in Figure 13, where the measured $M(H)$ dependence at 2 K and Brillouin function for $S_{\text{Cu}} = 1/2$ are compared. Brillouin function would simulate the strongly bound antiferromagnetic

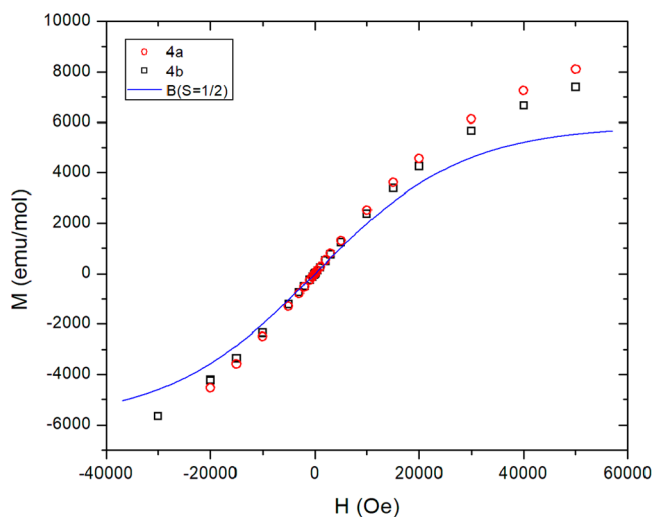


Figure 13. $M(H)$ measurements at 2 K for compounds **4a** and **4b**. The blue line is a simulated curve of the Brillouin function with spin $S = 1/2$.

Fe layers and isolated Cu magnetic centers. As shown in the Figure 13, magnetization is higher than expected for a single Cu paramagnetic ion but significantly lower than what would be the case for the ferromagnetic interaction between Fe^{2+} magnetic centers. From this, we can also confirm that the interaction between Fe^{2+} magnetic centers within honeycomb layers is antiferromagnetic and strong. Discrepancy between the measured $M(H)$ and the Brillouin function for $S_{\text{Cu}} = 1/2$ suggests that there is some kind of interaction between Cu^{2+} and Fe^{2+} ; they feel each other, although weakly. As already noted, DFT suggested ferromagnetic interaction of ~ 1 K between the Cu^{2+} ion and nearest Fe^{2+} ions. Also, the response of antiferromagnetic layer of iron(II) contributes slightly to the increase of magnetization with an increase of applied field, raising the measured $M(H)$ slightly above the Brillouin function. There is no magnetic hysteresis in $M(H)$ measurements, in accordance with the fact that the small ZFC–FC splitting is observable only in very small fields.

Low-temperature behavior is not typical one for pure antiferromagnetic state since the χT does not tend to zero as $T \rightarrow 0$, but rather extrapolates to the finite value between 2.6 and 2.7 (units of $N\mu_{\text{B}}^2/3k_{\text{B}}$), or eventually would make a plateau at around 3.2 that would correspond to independent Cu^{2+} ions, but this is not observed down to a temperature of 2 K. For the lowest temperatures, corresponding rise in susceptibility (inset in Figure 12) can be attributed to paramagnetic Cu^{2+} ions almost as independent spins (paramagnetic) or weakly bound to the Fe-planes (or even other Cu^{2+} ions), as suggested by DFT, calculated small superexchange value.

In conclusion, the antiferromagnetic transition at around 25 K (Figure 12) occurs due to relatively strong antiferromagnetic superexchange interactions in the honeycomb Fe^{2+} lattice and much weaker coupling between the layers coming from ferromagnetic superexchange transferred mainly through complex cation over $\text{O}_{\text{ox}}\text{—Cu—O}_{\text{ox}}$ bridges.

CONCLUSIONS

Results described in this paper show the great influence of simple ions (Cl^- and NO_3^-) of used copper(II) salts on the type of bridge and overall structural arrangements of investigated oxalate-based compounds reflecting on magnetic and electrical properties or on the occurrence of polymorphic forms.

Four novel heterometallic $[\text{CuFe}]$ compounds, polymorphs (a and b) of 3D $[\text{Cu}^{\text{II}}\text{Fe}^{\text{II}}_2(\text{H}_2\text{O})(\text{terpy})(\text{C}_2\text{O}_4)_3]_n$ (4) and 1D $\{[\text{Cu}^{\text{II}}_2\text{Fe}^{\text{III}}(\text{H}_2\text{O})(\text{terpy})_2(\text{C}_2\text{O}_4)_{7/2}] \cdot 6\text{H}_2\text{O}\}_n$ (3) coordination polymers, and complex salts $[\text{Cu}^{\text{II}}_4(\text{terpy})_4\text{Cl}_5] \cdot [\text{Fe}^{\text{III}}(\text{C}_2\text{O}_4)_3] \cdot 10\text{H}_2\text{O}$ (1) and $\{[\text{Cu}^{\text{II}}_2(\text{H}_2\text{O})_2(\text{terpy})_2(\text{C}_2\text{O}_4)] [\text{Cu}^{\text{II}}\text{Fe}^{\text{III}}(\text{CH}_3\text{OH})(\text{terpy})(\text{C}_2\text{O}_4)_3]_2\}$ (2), have been synthesized using the building block $[\text{Fe}^{\text{III}}(\text{C}_2\text{O}_4)_3]^{3-}$ and characterized.

These compounds show unique structures and related properties: (i) Depending on which copper salt $[\text{CuCl}_2 \cdot 2\text{H}_2\text{O}]$ or $[\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}]$ has been utilized in the methanol layer during liquid diffusion, chloro-bridged (1) or oxalate-bridged (2 and 3) compounds have been obtained. Also, the use of the chloride salt of copper(II) in the hydrothermal reactions produces a triclinic polymorph (4a), while a monoclinic polymorph (4b) has been formed with the nitrate salt. (ii) The partial decomposition of the tris(oxalate)ferate(III) anion during crystallization process of 2 and 3 has occurred. (iii) Uncommonly, compound 3 shows three different bridging modes of the oxalate group, while polymer

4 shows two. (iv) Compound 2 is the first structurally characterized compound in which Cu^{2+} and Fe^{3+} ions are bridged by bis(bidentate) oxalate group; the superexchange interaction between these ions is weakly antiferromagnetic. It appears much stronger within homodinuclear oxalate-bridged cation of copper(II) of the same compound. (v) Compound 1 is also interesting magnetically, since there is a chloro-bridged zigzag tetramer where two superexchange interactions are ferromagnetic and stronger and one is antiferromagnetic and weaker between copper(II) ions through chloride ions, making the ground state of this magnetic unit $S = 1$. (vi) Polymorphs 4a and 4b are the first magnetically and electrically investigated compounds in which the 2D honeycomb layers $[\text{Fe}_2(\text{C}_2\text{O}_4)_3]_n^{2n-}$ are bridged by cationic $[\text{Cu}(\text{H}_2\text{O})(\text{terpy})]^{2+}$ units forming 3D structural network of a novel topology ($12^3 \cdot 2 \cdot 12^5 \cdot 12 \cdot 3$). Magnetic phase transition appears at 25 K, and long-range order is established as a consequence of stronger antiferromagnetic interactions within the honeycomb layer and weaker interlayer interactions transferred via cation units. Also, they exhibit strong dependence of electrical conductivity on the relative humidity revealing proton conduction mechanism.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.inorgchem.0c02548>.

Experimental section, selected distances and angles for 1–4, ORTEP-3 drawing of 1–4, parameters of stacking, hydrogen bonding geometry and motifs, bridging modes of oxalate group in 3 and 4, stacking in 2, topological representation of 4a and 4b, TG and DTA curves, thermoanalytical data, PXRD pattern of sample obtained by heating 4a, selected IR bands, impedance plot for 2 and 4a, density of states, $\chi(T)$ of 4a and 4b, Curie–Weiss fits of 4a and 4b (PDF)

Accession Codes

CCDC 2017977–2017981 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: + 44 1223 336033.

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Notes

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